

ANILA IJAZ

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PROFESSIONAL EXPERIENCE

AI Research Engineer (Student Assistant) | TU Berlin (IWF) 03/2026 – Present

- Building modular ML pipelines for **time series analysis** of time-based industrial signals, applying anomaly detection to manufacturing process monitoring and reducing false positives by 20%.
- Developing **SHAP-based explainability (XAI)** layer, enabling non-technical stakeholders to interpret model decisions – driving client-facing communication of complex AI outputs.
- Driving ML projects from proof-of-concept through to production-ready prototypes, applying **version control, clean code, and CI/CD** practices throughout.
- Translating domain expert requirements into scalable software solutions through iterative collaboration – scoping feasibility and managing technical trade-offs across iterations.

AI Research Engineer (Student Assistant) | TU Berlin (SNET Labs) 05/2025 – 12/2025

- Implemented an **LLM-based agentic pipeline** from ideation to working prototype, reducing manual effort by 30% – scoping technical feasibility with stakeholders before each iteration.
- Integrated **REST APIs** for real-time data ingestion and **Natural Language Processing** tasks, improving downstream decision accuracy by 25%.
- Worked in a cross-functional, self-organised team; maintained **reproducible pipelines, unit testing**, and structured documentation aligned with sound engineering practices.
- Communicated findings clearly to both technical and non-technical stakeholders – designing and explaining complex AI solutions in accessible terms.

Software Engineer – Backend & Data Systems | NetSol Technologies, Pakistan 08/2022 – 09/2024

- Built and maintained backend services for enterprise financial clients, with a focus on **clean architecture**, data integrity, and long-term maintainability.
- Optimised SQL-based data pipelines, improving data access efficiency by 40% through query refactoring and structural improvements.
- Participated in **code reviews** and cross-team integration, contributing to shared engineering quality standards; collaborated with product owners to align implementation with client requirements.

Software Developer | Studio Mega Fury, Pakistan 06/2021 – 07/2022

- Developed modular backend components in **C#** and **C++**, applying design patterns and clean code principles for scalability.
- Participated in iterative development cycles including **code review**, debugging, and technical documentation – contributing to a shared culture of engineering quality.

SELECTED PROJECTS

Fleet Document Intelligence Platform [\[GitHub\]](#) *Python, FastAPI, LangChain, LangGraph, RAG, pgvector, Docker, Kubernetes, AWS EKS, CI/CD*

- End-to-end **production-grade AI automation** platform: ingests fleet leasing documents (invoices, contracts, damage reports), extracts structured data via **LLM pipelines**, and serves a **RAG chatbot** with cited answers over the full corpus.
- Architected a **LangGraph agentic triage pipeline** with n8n orchestration, **pgvector** semantic retrieval, React frontend, and full **Kubernetes deployment on AWS EKS** – driving the project from ideation through all 7 production phases.

- Applied **clean code**, **unit testing (16 tests)**, **CI/CD** (GitHub Actions), and multi-stage Docker builds; provider-agnostic LLM design supports OpenAI, Anthropic, AWS Bedrock, and Ollama.

InsightRAG – Production-Grade RAG Pipeline [GitHub] *Python, LangChain, FAISS, BM25, LLMs, FastAPI, Docker, CI/CD*

- Built a production-ready **RAG system** with hybrid retrieval (BGE + BM25 with RRF fusion), fine-tuned **cross-encoder reranker**, and prompt-injection guardrails – evaluated with **RAGAS** framework.
- Designed for enterprise Q&A over large document sets; containerised with **Docker** and integrated **CI/CD** pipelines for automated testing and deployment.

Intelligent Healthcare Chatbot [GitHub] *Python, LLMs, RAG, FastAPI*

- Designed and built a modular AI assistant using **RAG pipelines** and LLMs, with clean separation of concerns across retrieval, generation, and serving layers.
- Applied **MLOps practices** (FastAPI serving, structured logging, evaluation pipelines) to ensure the system was testable, maintainable, and extensible.

Real-Time Object & Shape Detection System [GitHub] *Python, OpenCV, YOLOv8*

- Implemented a real-time **Computer Vision** detection pipeline integrating deep learning (YOLOv8) with classical contour analysis; reduced false detection rate through iterative signal filtering.
- Built as independent, testable modules with preprocessing, filtering, and benchmarking components – applying clean architecture and open/closed design principles.

[Explore more projects on GitHub](#)

TECHNICAL SKILLS

Languages:	Python, C#, JavaScript, TypeScript, C++ , SQL
AI & ML:	PyTorch, TensorFlow, Scikit-learn, LLMs, RAG Pipelines, LangChain, LangGraph, Agentic AI, NLP, Computer Vision (YOLOv8, OpenCV), Anomaly Detection, Explainable AI (SHAP), MLflow
Engineering Practices:	Clean Code, Modular Architecture, Design Patterns, Code Review, Refactoring, Unit Testing, CI/CD, Automation, Agile/Scrum, Version Control
Cloud & Infra.:	AWS (EKS, ECR), Docker, Kubernetes, GitHub Actions, Git/GitHub, Linux/Bash, Jupyter
Web & APIs:	REST APIs, FastAPI, Flask, Node.js, React, HTML/CSS
Human Languages:	English (C1), German (A2 – actively improving), Urdu (Native)

EDUCATION

MSc Artificial Intelligence	Brandenburg University of Technology (BTU), Germany	10/2024 – 08/2026
BS Computer Science	University of Engineering & Technology (UET) Lahore, Pakistan	2017 – 2021

CERTIFICATIONS

Google Generative AI	LLMs, Prompt Engineering, AI Applications – Badge Awarded	2024
MERN Stack Development	Full-Stack Web Development Certification	2022
Image Processing Workshop	Certificate of Excellence in Image Processing & Computer Vision	2019

EXTRACURRICULAR & VOLUNTEER EXPERIENCE

- **Coordinator, ProSoft Society** – Organised tech events, workshops, and programming competitions for the CS department at UET Lahore.